

POLITECNICO
MILANO 1863

COMPUTER SCIENCE AND ENGINEERING
SOFTWARE ENGINEERING II
2025 - 2026

ATD

Acceptance Test Deriverable

Software:

Best Bike Paths (BBP)

Team:
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1| Introduction and Scope

This document has the purpose of showing the results of the acceptance tests performed on the *Best Bike Path* platform developed by another team.
The tests are performed on the system to verify that the system meets the requirements and the expectations of the customer.
The following sections provide a complete and detailed description of the test cases performed according to their R&DD and the assignments, the results obtained, and any bugs, issues, or incomplete functionalities identified during the testing phase.

1.1 Tested Project

The project analyzed was developed by the team composed by:

- Youssef Mohamed Elsayed Maati Rabie
- Arsenii Prykhodko

Team's repository: https://github.com/YoussefMoRabie/PrykhodkoNataliRabie

1.2 Scope

The scope of the testing activity is limited to the features assigned to individual students during the Requirements and Design Document (R&DD) phase and subsequently implemented by the reviewed team, which consists of two members, during the Implementation phase.
In particular, the system should allow user to:

1. Record personal trips;
2. Insert publishable information in manual mode;
3. Visualize information about possible paths between an origin and a destination specified by the user.

2| Installation Setup

The installation process is smooth, and the README provided by the team is clear and easy to follow. All necessary materials are included in the APK package, so the only step required is to download and install it. During installation, you will need to grant permission when prompted by Google Play Protect, which is also clearly explained in the README. Overall, the setup process is seamless and intuitive.

When using the app, the phone's GPS must be enabled. If it is turned off, the app will prompt you to activate it, ensuring that users are aware of the requirement. Therefore, there are no issues from either the installation or the initial setup perspective.

3| Acceptance Test Cases

The test cases are taken entirely from the use cases reported in the RASD document.

Name	Registration on BBP (Best Bike Paths) platform
Input	The user enters valid credentials (email, password, confirmpassword) and clicks the registration button.
Expected	The user is successfully registered on the BBP platform and can access the main dashboard.
Pass/Fail	Pass
Note	Registration functionality works correctly with no detected issues.

Name	Login to BBP (Best Bike Paths) platform
Input	The registered user enters correct login credentials and clicks the login button.
Expected	The user is successfully logged into the BBP platform and redirected to the dashboard.
Pass/Fail	Pass
Note	Login functionality operates normally without errors. In case of failure credentials it shows correct indications.

Name	Record Trip on BBP (Best Bike Paths) platform
Input	A registered user logs into the BBP platform, starts a trip recording, rides, and then stops the recording.
Expected	Only logged-in users can record trips. After the trip ends, the system saves the trip and displays statistics such as distance and duration.
Pass/Fail	Pass
Note	Trip recording works correctly. Access is properly restricted to registered users, and trip data is shown after completion in the user profile. But, the recorded path becomes visible only after the record is completed, meaning that when recording, the recording path is not shown on the screen.

Name	Manual path information insertion on BBP platform
Input	A logged-in user manually enters path information, including path names, path status, and obstacle details, then submits the form.
Expected	The system successfully stores the manually entered information and updates the path rating accordingly.
Pass/Fail	Pass
Note	Manual data insertion functions correctly, and path evaluations reflect the submitted information.

Name	Weather data recording after trip completion on BBP platform
Input	A logged-in user records a bike trip and ends the ride.
Expected	The system retrieves and stores the weather data at the time of the trip, including temperature and wind conditions, and displays this information in the trip details.
Pass/Fail	Pass
Note	Weather information is accurately recorded and shown after the trip ends.

Name	Search Bike Paths on BBP (Best Bike Paths) platform
Input	The user enters an origin and a destination to search for available bike routes.
Expected	The system should display all feasible routes that users can ride between the selected points.
Pass/Fail	pass
Note	The platform currently shows only dedicated bike lanes in Milan instead of all rideable roads. This limits route availability and may affect navigation accuracy.

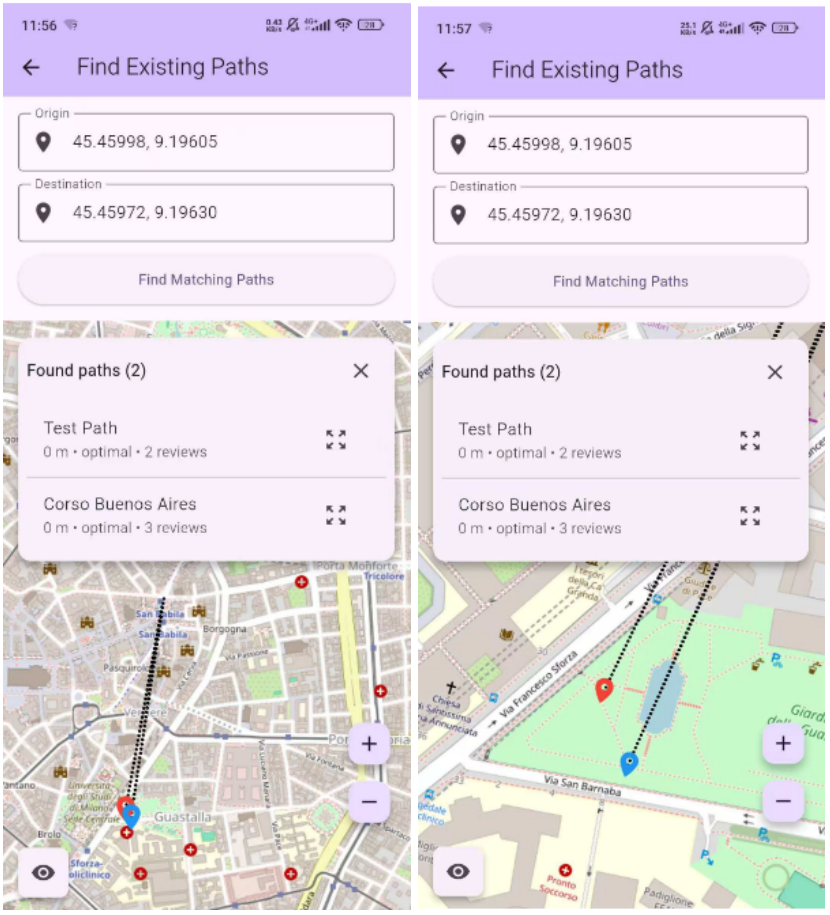
3.1 More about the use cases

Features outside the assigned scope or not implemented by the reviewed group are not considered in this document.

According to the assignment, a bike path should either be a dedicated cycling track or a road with minimal car traffic and speed limits compatible with the average speed of a bicycle. However, the app defines only a very limited number of dedicated bike paths, which reduces its relevance and overall usefulness for users.

For users located away from these predefined routes, the search results first direct them to join one of the preset paths before guiding them toward their destination. In some cases, the time required to detour to a predefined path is significantly longer than taking a more direct route. Instead of generating a convenient and customized path from the user's actual location, the app prioritizes these preset routes, even when they are inefficient. This design choice may negatively affect usability, route efficiency, and the overall user experience.

Some of the routes found are not real paths; they appear as straight lines, causing them to pass through areas such as rooftops or bodies of water (see screenshot on next page).

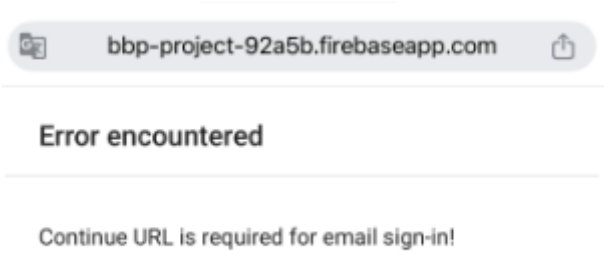


Screenshots of Route Search Function

4| Additional comment

The developers have added a new function to the project that allows users to register using only their email address. However, this feature did not work during the testing session. After clicking “Sign in with email” and entering an email address, the user receives an email containing a link. However, the link redirects to an error page stating that “Continue URL is required for email sign-in!” .

Nevertheless, this function is not a requirement of the assignment, and the standard registration and login method—registering with an email address and password, then logging in with those credentials—works properly. Therefore, I would not consider this a significant issue.



Error page directed by the email Link

5| Conclusion

The app features a clean and minimalist interface, which makes both the installation and overall user experience straightforward and intuitive. Users can quickly understand how to navigate the app without requiring additional guidance, reflecting thoughtful design and usability considerations. Moreover, the installation process is smooth, allowing users to set up the app with minimal effort.

The application satisfies all the requirements of the assignment and remains consistent with the specifications outlined in the RASD and DD documents. The implemented features align well with the proposed design, demonstrating a clear effort to translate the project documentation into a functional product.

Although there are some minor issues in the implementation and a few details that could be further refined, these do not significantly affect the overall functionality of the app. Considering its current stage of development, the project successfully fulfills its role as a prototype. Overall, it represents a well-executed and high-quality prototype with strong potential for further improvement and expansion.

6| Effort Spent

Name	Time spent
Jiaxin Yang	2h
Run Jie Simone Dai	2h
Jiayi Zheng	2h

7| References

1. *GitHub Ripository*. (n.d.). https://github.com/YoussefMoRabie/PrykhodkoNataliRabie
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